Ex.no:7A	DATA VISUALIZATION USING TABLEAU – STUDENT PERFORMANCE

Aim:

To connect, analyze, visualize, and create interactive dashboards using Tableau for effective business intelligence and data analytics.

Algorithm:

Algorithm: Student Performance Visualization using Tableau

Step 1: Data Loading

- Connect the synthetic Student Performance dataset to Tableau.
- Verify the imported fields: Student, Department, and Marks.
- The dataset contains six students: Arun, Bala, Divya, Priya, Ravi, and Karthik.

Step 2: Student Marks Bar Chart

- Create a worksheet for student-wise marks.
- Place Student on Columns.
- Place Marks on Rows.
- Select the Bar Chart visualization.
- Enable data labels to display the marks of each student.

Step 3: Department-wise Marks Visualization

- Create a worksheet using Department and Marks.
- Aggregate Marks using SUM.
- Visualize the contribution of each department using a Pie Chart.
- Display department names and corresponding marks as labels.

Step 4: Student Performance Trend

- Create a Line Chart using Student and Marks.
- Place Student on Columns and Marks on Rows.
- Enable labels to display individual marks.
- Use the chart to compare the performance of students.

Step 5: Marks Percentage

- Create a calculated field named Percentage.

- Use the formula:
  [Marks]
- Display Student, Marks, and Percentage in a worksheet.

Step 6: Department Filter

- Add Department to the Filters shelf.
- Select a department such as AI&DS.
- Observe the updated student marks based on the selected department.

Step 7: Minimum Marks Parameter

- Create a parameter named Minimum Marks.
- Set the required minimum marks value.
- Apply the parameter to display only students whose marks meet or exceed the selected value.
- For example, when Minimum Marks = 85, display students scoring 85 or above.

Step 8: Dashboard Creation

- Create a new Tableau Dashboard.
- Add the Student Marks, Department Contribution, Marks Percentage, Student Performance Trend, and Minimum Marks visualizations.
- Arrange the worksheets clearly.
- Add the Department filter and Minimum Marks parameter.
- Format the dashboard with suitable titles and labels.

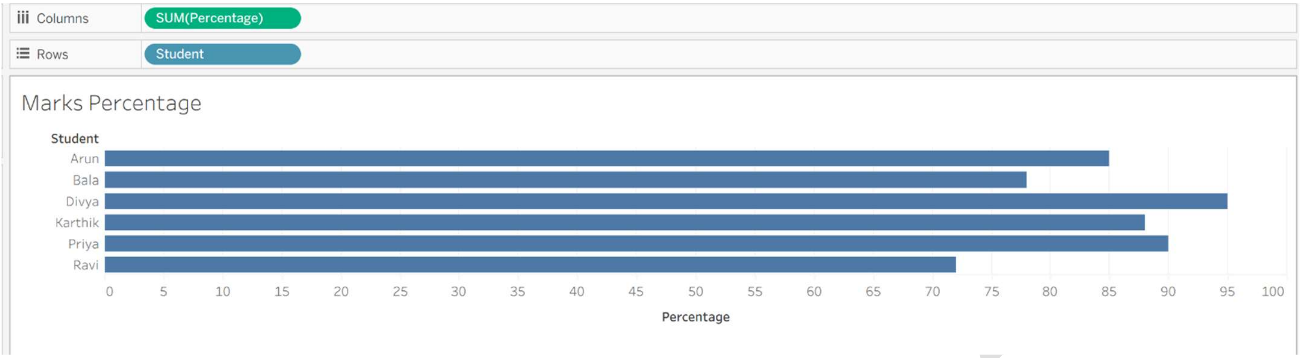
Step 9: Story Creation

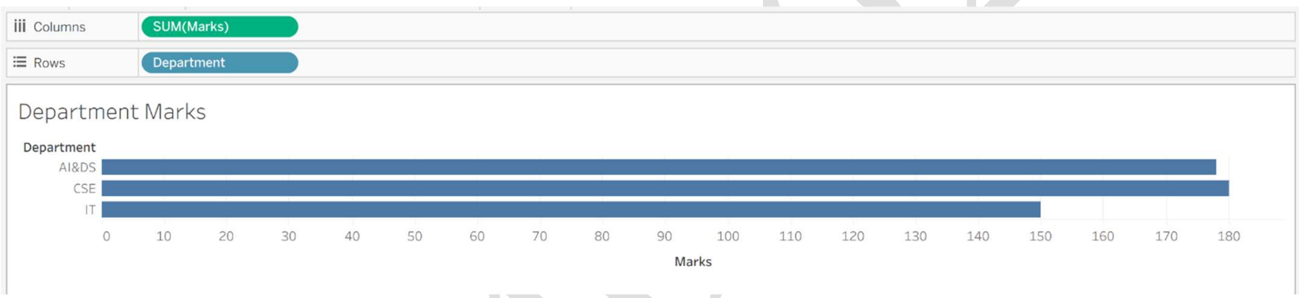
- Create a Tableau Story using the completed dashboard.
- Add story points for:
    1. Overall Student Performance
    2. Department-wise Analysis
    3. Top Performing Students
    4. Final Performance Analysis
- Present the visual insights using the story.

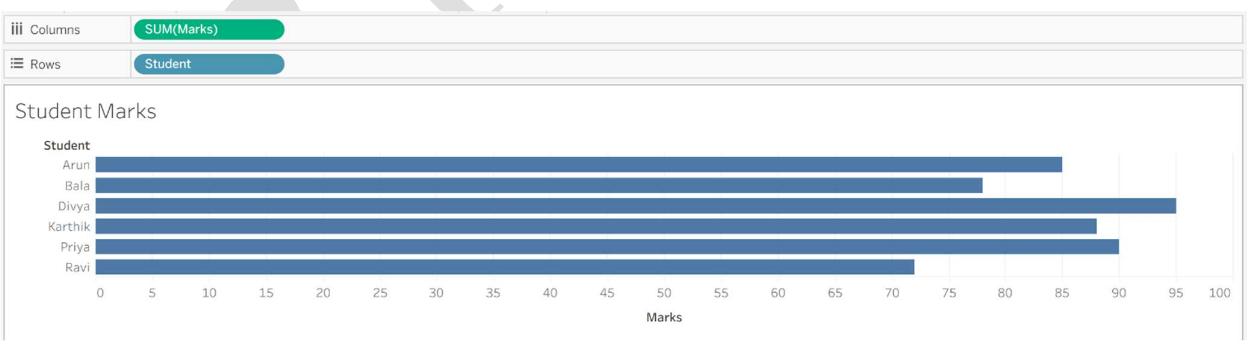
Step 10: Analysis

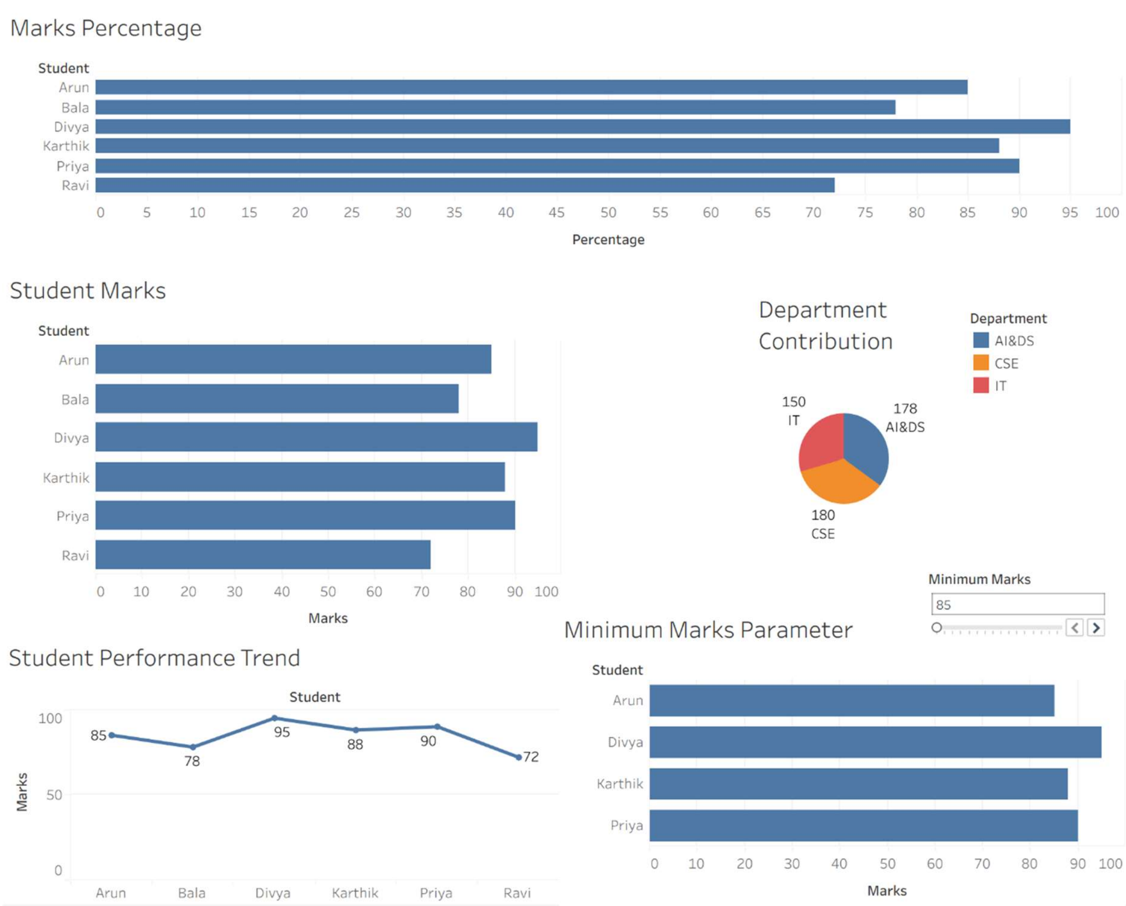
- Analyze the charts to identify the highest and lowest performing students.
- Compare marks among departments.
- Observe how the Department filter and Minimum Marks parameter dynamically change the visualizations.
- Use the dashboard and story to summarize the overall student performance.

Output:


Columns
SUM(Percentage)
Rows
Student
Marks Percentage
Student
Arun
Bala
Divya
Karthik
Priya
Ravi
0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
Percentage


Columns
SUM(Marks)
Rows
Department
Department Marks
Department
AI&DS
CSE
IT
0 10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160 170 180
Marks


Columns
SUM(Marks)
Rows
Student
Student Marks
Student
Arun
Bala
Divya
Karthik
Priya
Ravi
0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
Marks


Marks Percentage
Student
Arun
Bala
Divya
Karthik
Priya
Ravi
0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
Percentage
Student Marks
Student
Arun
Bala
Divya
Karthik
Priya
Ravi
0 10 20 30 40 50 60 70 80 90 100
Marks
Department Contribution
Department
AI&DS
CSE
IT
150 IT
178 AI&DS
180 CSE
Minimum Marks
85
Minimum Marks Parameter
Student
Arun
Divya
Karthik
Priya
0 10 20 30 40 50 60 70 80 90 100
Marks
Student Performance Trend
Student
100
50
0
Marks
85
78
95
88
90
72
Arun Bala Divya Karthik Priya Ravi

Result: